

BIS CIVIL ENGINEERING DEPARTMENT (CED)

RAG / PDF Ingestion Knowledge Base — Civil Engineering, Building Construction, Materials & Structural Safety

Purpose: Retrieval-oriented reference for an AI assistant answering questions about BIS Civil Engineering standards. It covers the department scope, technical committees, major standard families, important standards and current-status cautions. It is not a substitute for purchasing or consulting the complete official standard text.

Source basis: BIS Civil Engineering Department / CEDC materials and current BIS Standards records retrieved in September 2026.

1. Department Scope

The Civil Engineering Department (CED) covers standardization in civil engineering, including structural engineering, building materials and components, planning, design, construction and maintenance of civil engineering structures and the built environment, construction practices, and safety in buildings. Its scope excludes subjects specifically related to Water Resources Development and Management.

CED works through the Civil Engineering Division Council (CEDC) and sectional technical committees. The BIS committee framework covers cement/concrete, sanitary appliances and water fittings, building limes and gypsum, flooring/wall finishing/roofing, stones, structural engineering, timber, doors/windows, functional building requirements, construction practices, builders' hardware, wood/lignocellulosic products, fire fighting, public health engineering, construction management, clay/stabilized products, furniture, fire safety, structural safety, special structures, earthquake engineering, waterproofing/damp-proofing, soil/foundations, measurement of works, National Building Code, and other civil-engineering subjects.

2. CED Technical Committee Map

Committee	Name	Main subject area
CED 02	Cement and Concrete	Cement, concrete, concrete materials, mix proportioning, testing and related products.
CED 03	Sanitary Appliances and Water Fittings	Sanitary appliances, plumbing and water fittings.
CED 04	Building Limes and Gypsum Products	Building lime, gypsum and related construction products.
CED 05	Flooring, Wall Finishing and Roofing	Flooring systems, wall finishes, roofing materials and related construction products.
CED 06	Stones	Natural stones and stone products used in construction.
CED 07	Structural Engineering and Structural Sections	Structural engineering using steel/aluminium structural sections, including structural products.
CED 09	Timber and Timber Stores	Timber, timber preservation/storage and related requirements.
CED 11	Doors, Windows and Shutters	Doors, windows, shutters and associated construction requirements.
CED 12	Functional Requirements in Buildings	Functional requirements and performance-oriented requirements for buildings.
CED 13	Building Construction Practices	Building construction practices, masonry, workmanship and associated methods.
CED 15	Builders' Hardware	Hardware used in building construction.
CED 20	Wood and Other Lignocellulosic Products	Wood-based and other lignocellulosic construction/products.
CED 22	Fire Fighting	Fire-fighting equipment and systems.
CED 24	Public Health Engineering	Public-health engineering services and building sanitation-related engineering.
CED 29	Construction Management (Including Safety in Construction)	Construction management, site safety and construction safety.
CED 30	Clay and Stabilized Products for Construction	Clay products, stabilized earth and related construction materials.
CED 35	Furniture	Furniture-related standardization.
CED 36	Fire Safety	Fire safety of buildings and related requirements.
CED 37	Structural Safety	Structural safety subjects.
CED 38	Special Structures	Special structures, including specialized structural systems.
CED 39	Earthquake Engineering	Earthquake-resistant design and seismic engineering.
CED 41	Water-Proofing and Damp-Proofing	Waterproofing, damp-proofing and protection against water ingress.
CED 43	Soil and Foundation Engineering	Soil engineering, subsurface investigation and foundation design/construction.

Committee	Name	Main subject area
CED 44	Methods of Measurement of Works of Civil Engineering	Measurement methods for building/civil engineering works, excluding water-resources devel
CED 46	National Building Code	National Building Code and its associated provisions/standards.

3. High-Priority Concrete and Cement Standards

Standard	Title	Use in retrieval	Committee
IS 456:2000	Plain and reinforced concrete — Code of practice (Fourth Revision)	Code of plain-concrete design and construction code. BIS records show review in 2025 and	CED 02
IS 1343:2012	Prestressed concrete — Code of practice (Second Revision)	Prestressed concrete design and construction.	CED 02
IS 10262:2019	Concrete mix proportioning — Guidelines (Second Revision)	Guidelines for concrete mix proportioning.	CED 02
IS 383:2016	Coarse and fine aggregate for concrete — Specification (Third Revision)	Requirements for concrete.	CED 02
IS 9103:1999	Concrete admixtures — Specification	Requirements for admixtures used in concrete.	CED 02
IS 1199 series	Methods of sampling and analysis/testing of concrete	Concrete sampling and testing methods; use the exact current	CED 02 when answering a
IS 516 series	Methods of tests for strength of concrete	Concrete strength testing, including relevant mechanical test methods	CED 02
IS 1786:2008	High strength deformed steel bars and wires for concrete reinforcement — Specification (Fourth Revision)	Reinforcement steel specification.	CED 02 / related reinforcement

4. Structural Steel, Loads and Structural Design

Standard	Title	AI retrieval meaning
IS 800:2007	General construction in steel — Code of practice	Primary Indian code for structural steel design and construction.
IS 875 (Part 1)	Design loads (other than earthquake) — Dead loads	Dead loads for buildings/structures.
IS 875 (Part 2)	Design loads — Imposed loads	Imposed/live loads.
IS 875 (Part 3)	Design loads — Wind loads	Wind loading provisions.
IS 875 (Part 4)	Design loads — Snow loads	Snow loading provisions where applicable.
IS 875 (Part 5)	Design loads — Special loads and load combinations	Special loads and load combinations.
IS 2062:2025 (Part 1)	Structural steel — Part 1 — Hot rolled medium and high tensile steel	Structural steel product specification; current BIS records show the 2025 edition
IS 7215:1974	Tolerances for fabrication of steel structures	Fabrication tolerances; BIS record reviewed in 2026.
IS 808	Dimensions for hot rolled steel beam, column, channel and angles	Standardized dimensions for hot-rolled structural sections; retrieve the current e

5. Earthquake and Seismic Engineering

Critical currency warning: BIS currently lists IS 1893 (Part 1):2025 as **withdrawn on 05 March 2026**. Therefore an AI assistant must not simply answer that IS 1893 (Part 1):2025 is the current applicable edition. The current BIS record must be checked before giving compliance guidance.

Standard	Title	Retrieval guidance
IS 1893 (Part 1):2025	Design Earthquake Hazard and Criteria for Earthquake-Resistant Buildings	Design of structures marked as General Provisions (95-Mar-2026) is superseded by IS 1893 (Part 1):2025.
IS 1893 (Part 2):2014	Criteria for Earthquake Resistant Design of Structures — Part 2: Seismic Design of Tanks	Seismic design of tank-retaining tanks; reviewed in 2024 with Amendment 1.
IS 4326:2013	Earthquake resistant design and construction of buildings — Code of practice	Building design/construction practices for earthquake resistance; CED 39.
IS 13920:2016	Ductile design and detailing of reinforced concrete structures subjected to seismic forces	Code of practice for detailing; verify current BIS lifecycle before compliance use.
IS 13827:1993	Improving earthquake resistance of earthen buildings — Guidelines	Guidelines for improving seismic resistance of earthen buildings.
IS 13828:1993	Improving earthquake resistance of low-strength masonry buildings	Guidelines for low-strength masonry buildings in seismic regions.

6. Soil, Foundation and Geotechnical Engineering

Standard/family	Title	Use
IS 1904:2021	General requirements for design and construction of foundations in soils	General all practice (Fourth Edition) BIS record reviewed in 2026.
IS 1080:1985	Code of practice for design and construction of shallow foundations in soils	Shallow foundations (and shell)
IS 1892:2021	Subsurface investigation for foundations — Code of practice (Second Revision)	Soil subsurface investigation for foundations; reviewed in 2026.
IS 2911 series	Design and construction of pile foundations — Code of practice	Pile foundations, with separate parts/sections for different pile types
IS 2131:2025	Standard penetration test of soil — Method of test	SPT method; current BIS records show review in 2025.
IS 2720 series	Methods of test for soils	Large family of soil-test methods; exact part must be identified for a
IS 4091:1979	Code of practice for design and construction of foundations for transmission lines	Specialised code for transmission-line towers/poles.

7. Building Construction, Masonry and Materials

Standard	Title	AI retrieval meaning
SP 7:2026	National Building Construction Standards 2026	Current BIS record lists SP 7:2026 as superseding SP 7:2016 and identifying the new standards.
IS 1905:1987	Structural use of unreinforced masonry — Code of practice (Third Revision)	Design and use of unreinforced masonry; BIS record shows references to current standards.
IS 1077:2025	Common burnt clay building bricks — Specification (Sixth Revision)	Building-brick product specification; use current BIS record for exact requirements.
IS 2185 (Part 1):2005	Concrete masonry units — Specification: Part 1 Hollow and solid concrete blocks	Concrete block specification.
IS 6041:1985	Code of practice for construction of autoclaved cellular concrete blocks	Autoclaved cellular concrete construction.
IS 2502:1963	Code of practice for bending and fixing of bars for concrete reinforcement	Reinforcement bar bending/fixing.
IS 4082:1996	Stacking and storage of construction materials and components at construction sites	Construction site storage (Sixth Revision)
SP 62:1997	Handbook on causes and prevention of cracks in buildings	Reference handbook for building cracks and prevention.

8. Fire Safety, Public Health, Waterproofing and Construction Management

Domain	Committee	Retrieval guidance
Fire Safety	CED 22 / CED 36	Fire-fighting equipment and building fire safety. Relevant building fire-safety standards should be consulted.
Public Health Engineering	CED 24	Sanitation, plumbing and public-health engineering services in buildings.
Waterproofing and Damp-proofing	CED 41	Waterproofing and damp-proofing systems, materials and construction practices.
Construction Management & Safety	CED 29	Construction management and safety in construction, including safe work practices.
Methods of Measurement	CED 44	Measurement of building/civil engineering works; exact part of IS 1200 or successor/current standards.
National Building Code	CED 46	National Building Code framework. The BIS current record lists SP 7:2026 as National Building Construction Standards.

9. High-Value Questions for the BIS AI Assistant

What is CED in BIS?

CED is the Civil Engineering Department of the Bureau of Indian Standards. It handles standardization across civil engineering, structural engineering, building materials/components, planning, design, construction, maintenance, construction practices and building safety.

Which BIS standard is the main code for reinforced concrete?

IS 456:2000 is the principal code for plain and reinforced concrete. The BIS record shows it was reviewed in 2025 and has amendments, including Amendment 6 in 2024. Exact design requirements should be taken from the current official standard.

Which BIS standard is used for prestressed concrete?

IS 1343:2012 — Prestressed Concrete — Code of Practice.

What is IS 10262?

IS 10262:2019 provides guidelines for concrete mix proportioning.

What is IS 800?

IS 800:2007 is the general code of practice for construction in steel.

Which IS 875 part covers wind loads?

IS 875 (Part 3) covers wind loads. The exact current edition/status should be checked in BIS before compliance use.

Which committee handles structural engineering?

CED 07 — Structural Engineering and Structural Sections.

Which committee handles earthquake engineering?

CED 39 — Earthquake Engineering.

Which committee handles soil and foundations?

CED 43 — Soil and Foundation Engineering.

Which committee handles the National Building Code?

CED 46 — National Building Code.

Which standard covers general foundation design and construction in soils?

IS 1904:2021 — General requirements for design and construction of foundations in soils — Code of practice.

Which standard covers subsurface investigation for foundations?

IS 1892:2021 — Subsurface investigation for foundations — Code of practice.

What is SP 7:2026?

BIS currently lists SP 7:2026 as National Building Construction Standards 2026, superseding SP 7:2016, under CED 46.

Is IS 1893 (Part 1):2025 current?

The current BIS record retrieved for this knowledge base says IS 1893 (Part 1):2025 was withdrawn on 05 March 2026. Therefore the assistant must check the current BIS lifecycle rather than presenting the 2025 edition as automatically current.

10. AI Retrieval and Compliance Rules

- **Never equate an Indian Standard with mandatory certification.** Many civil-engineering standards are codes, methods, specifications or voluntary standards. Certification status must be taken from the current BIS record or applicable regulation.

- **Always preserve the full standard designation.** For example, distinguish IS 1893 (Part 1), IS 1893 (Part 2), IS 2911 parts/sections, and individual IS 2720 test parts.
- **For exact values, use the official standard text.** Do not invent load factors, concrete mix limits, reinforcement limits, soil parameters, test conditions, tolerances or acceptance criteria.
- **Check lifecycle status.** Standards may be revised, amended, reaffirmed, withdrawn or superseded. The assistant should retrieve the latest BIS record before answering compliance questions.
- **Ask for missing context.** For a building/design question, identify structure type, material, occupancy/use, location/seismic context where relevant, design stage, and whether the user wants design guidance, testing, product specification or certification information.
- **Separate standard categories.** Product specifications define products; codes of practice address design/construction; methods of test define testing; handbooks provide guidance; the National Building Code provides a broader building regulatory/code framework.
- **Use source-backed answers.** When possible, return the standard number, title, committee, status, and official BIS record/reference instead of a generic paragraph.

11. Currency and Source Notes

This PDF was prepared using official BIS sources retrieved in September 2026. The BIS CED overview identifies the department and its technical committees; BIS Standards records were used to verify selected standard titles, committees and lifecycle/review information. The current record for SP 7:2026 identifies it as National Building Construction Standards 2026 under CED 46. The current record for IS 456:2000 shows review in 2025 and amendments through 2024. The current record for IS 1893 (Part 1):2025 reports withdrawal on 05 March 2026. These examples demonstrate why the AI system should check live/current BIS metadata for compliance questions.

Recommended ingestion keywords: CED, CEDC, Civil Engineering Department, cement, concrete, RCC, prestressed concrete, structural steel, IS 800, IS 875, IS 456, IS 10262, earthquake, IS 1893, IS 4326, foundation, soil, IS 1904, IS 1892, pile foundation, IS 2911, NBC, SP 7, masonry, bricks, fire safety, waterproofing, construction safety, measurement of civil works.

End of CED Knowledge Base